

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Cash flows from operating activities for the year ended 31 December 2019

	HK\$'000	HK\$'000
Net income (W1)		1,770
Depreciation		462
Change in current assets and current liabilities		
Increase in accounts receivables	(2,420)	
Increase in inventory	(2,928)	
Increase in accounts payables	1,430	
Increase in accruals	400	
		<u>(3,518)</u>
Cash flow from operating activities		<u>(1,286)</u>
(W1) Revenue		12,450
Less: Cost and operating expenses		<u>10,680</u>
Net income		<u>1,770</u>

Answer 1(b)

From Answer 1(a) above, the cash flow generated from operating activities of Timberline was in deficit.

More than double increases in both accounts receivables and inventory far outweigh the 25% increase in sales revenue. This indicates that Timberline may have trouble collecting its trade debts from its customers and may likely invest adversely in inventory.

The fact that accounts payables have also jumped drastically indicates that Timberline may have difficulty in settling its trade bills on time. The significant increase signifies that the current operating cash flow position of Timberline may be at the alarming stage.

Increase in accruals also echoes the implications above.

Answer 1(c)

Free cash flow is the net cash flow generated by operating activities minus capital expenditures during the year.

Free cash flow for Timberline in the financial year 2019

	HK\$'000
Net cash from operating activities	(1,286)
Capital expenditures	(3,400)
Free cash flow	<u>(4,686)</u>

The negative amount of net cash generated by operating activities during the year cannot meet the financial requirement of maintaining Timberline's non-current assets investments. Timberline had to resort to fund raising by issuing debts and/ or shares to finance the mentioned capital expenditure during the year.

Answer 2(a)(i)

40% of the heritance has been invested in ICDC shares. The question is how to invest the remaining 60% of the heritance to achieve the highest possible expected return. Since HK Mobile offers an expected return of 28%, higher than that offered by CK Property, the cash amount of HK\$600,000 should be invested in HK Mobile. The expected return with this portfolio is:

$$0.4 \times 18\% + 0.6 \times 28\% = 24\%$$

Answer 2(a)(ii)

CK Property has a lower risk as measured by its standard deviation, but has a much higher correlation coefficient with ICDC. To identify a portfolio that gives rise to the lowest level of risk, we need to compute the portfolio risks for both combinations: ICDC and CK Property, and ICDC and HK Mobile, as follows:

$$\begin{aligned}\sigma_{\text{ICDC-CK Property}} &= [0.4^2 \times 0.23^2 + 0.6^2 \times 0.34^2 + 2 \times 0.6 \times 0.4 \times 0.8 \times 0.23 \times 0.34]^{0.5} \\ &= [0.008464 + 0.041616 + 0.0300288]^{0.5} \\ &= 0.0801088^{0.5} = 0.283035 \\ \sigma_{\text{ICDC-HK Mobile}} &= [0.4^2 \times 0.23^2 + 0.6^2 \times 0.48^2 + 2 \times 0.6 \times 0.4 \times 0.3 \times 0.23 \times 0.48]^{0.5} \\ &= [0.008464 + 0.082944 + 0.0158976]^{0.5} \\ &= 0.1073056^{0.5} = 0.327575\end{aligned}$$

Therefore, the ICDC-CK Property portfolio is the lowest risk since it has the lowest standard deviation.

Answer 2(b)(i)

The ultimate goal for financial management is to maximise the market share value of a firm.

Financial managers make decisions affecting the current market share prices, which are determined by the future cash flows a firm can generate.

Managers can affect cash flows generated in different ways. They can, for example, select products or services to produce, assets to purchase and advertising campaigns to undertake, etc.

Answer 2(b)(ii)

Short-term debt instruments with maturities of less than a year are traded in money markets, while capital markets are where equity and debt instruments with maturities exceeding one year are traded.

Answer 2(b)(iii)

Initial public offering ("IPO") is the process of offering shares in a private company to the general public for the first time and being traded in the primary market.

After IPO, securities are traded by investors in the secondary market. When the public trades stocks on the Hong Kong Stock Exchange, it is engaging in trades in the secondary market.

Investors are more favourable to trade securities for which there is an active secondary market. Companies whose securities have active secondary markets have comparatively lower funding costs than similar companies whose securities do not have active secondary markets.

Answer 3(a)

$$R_E = 3\% + 1.6 (12\% - 3\%) = 17.4\%$$

$$R_P = \text{HK\$}100 \times 6\% / \text{HK\$}94 = 6.38\%$$

$$P_0 = \text{HK\$}1,000 / (1 + R_D / 2)^{20} = \text{HK\$}620$$

$$R_D = 4.84\%$$

Answer 3(b)

$$E = 50 \text{ million} \times \text{HK\$}12 = \text{HK\$}600 \text{ million}$$

$$P = 8 \text{ million} \times \text{HK\$}94 = \text{HK\$}752 \text{ million}$$

$$D = 7 \text{ million} \times \text{HK\$}620 = \text{HK\$}4,340 \text{ million}$$

$$V = \text{HK\$}600 \text{ million} + \text{HK\$}752 \text{ million} + \text{HK\$}4,340 \text{ million} = \text{HK\$}5,692 \text{ million}$$

$$W_E = E/V = \text{HK\$}600 \text{ million} / \text{HK\$}5,692 \text{ million} = 10.54\%$$

$$W_P = P/V = \text{HK\$}752 \text{ million} / \text{HK\$}5,692 \text{ million} = 13.21\%$$

$$W_D = D/V = \text{HK\$}4,340 \text{ million} / \text{HK\$}5,692 \text{ million} = 76.25\%$$

$$\text{WACC} = 10.54\% \times 17.4\% + 13.21\% \times 6.38\% + 76.25\% \times 4.84\% \times (1 - 20\%) = 5.63\%$$

Answer 3(c)

$$\begin{aligned}\text{NPV(A)} &= -\text{HK\$1,280,000} + \text{HK\$289,000} / 1.0563 + \text{HK\$336,000} / 1.0563^2 + \text{HK\$570,800} / 1.0563^3 \\ &\quad + \text{HK\$823,400} / 1.0563^4 \\ &= \text{HK\$440,438.74}\end{aligned}$$

$$\begin{aligned}\text{NPV(B)} &= -\text{HK\$738,000} + \text{HK\$138,000} / 1.0563 + \text{HK\$258,000} / 1.0563^2 + \text{HK\$301,000} / 1.0563^3 \\ &\quad + \text{HK\$598,000} / 1.0563^4 \\ &= \text{HK\$359,609.33}\end{aligned}$$

RRC should choose Project A because it has a higher NPV.

Answer 3(d)

$$\begin{aligned}\text{PI(A)} &= \frac{(\text{HK\$289,000} / 1.0563 + \text{HK\$336,000} / 1.0563^2 + \text{HK\$570,800} / 1.0563^3 + \text{HK\$823,400} / 1.0563^4)}{\text{HK\$1,280,000}} \\ &= 1.34 \\ \text{PI(B)} &= \frac{(\text{HK\$138,000} / 1.0563 + \text{HK\$258,000} / 1.0563^2 + \text{HK\$301,000} / 1.0563^3 + \text{HK\$598,000} / 1.0563^4)}{\text{HK\$738,000}} \\ &= 1.49\end{aligned}$$

RRC should choose Project B because it has a higher PI.

Answer 3(e)

RRC should choose Project A with a higher NPV.

NPV is the most preferable capital budgeting appraisal technique because it has no ranking problem and it can rank mutually exclusive projects.

Answer 4(a)

Segment reporting refers to the concept of reporting the operations of specific segments of a company in the disclosures that go along with financial statements.

Publicly held entities are required to undertake segment reporting but privately held companies are not.

The idea behind segment reporting is to provide investors and creditors with more information transparency regarding the financial results and the financial position of various and more significant operating units of a company for their relevant decision making.

A summary of the entire company does not reveal the details of how each segment operated, so segment reporting provides stakeholders with a better sense of the fluctuations of each segment to better monitor the overall corporate's management performance.

In case of higher earnings than expected, segment reporting also enables financial statement users to better identify where those earnings are coming from with more worthwhile insights to appraise the growth and sustainability of an entity.

Answer 4(b)

The segmented statement of profit or loss is as follows:

	<u>Badminton</u>	<u>Tennis</u>	<u>Company</u>
	<u>Rackets</u>	<u>Rackets</u>	
	HK\$	HK\$	HK\$
Sales	18,120,000	27,720,000	45,840,000
Variable cost	<u>7,550,000</u>	<u>12,504,800</u>	<u>20,054,800</u>
Contribution margin	10,570,000	15,215,200	25,785,200
Fixed cost	<u>6,500,000</u>	<u>9,850,000</u>	<u>16,350,000</u>
Segment margin	4,070,000	5,365,200	9,435,200
Corporate fixed cost			<u>8,080,000</u>
Net Profit			<u>1,355,200</u>

Answer 4(c)

There are four basic types of responsibility centres: cost centres, revenue centres, profit centres and investment centres. These centres indicate the degree of responsibility the manager has for performance and accountability.

Managers of cost centres are evaluated on their ability to control costs and expenses through budgeting and budgetary control.

Managers of revenue centres are evaluated by meeting revenue forecasts while controlling related selling costs.

Managers of profit centres are accountable for both revenue and cost, hence are judged on the level of profitability of their centres.

Lastly, managers of investment centres are evaluated on both the profitability of the centres and the rate of return earned on the funds invested.

The Badminton Rackets segment belongs to profit centre because it incurs costs and expenses, and generates revenue.

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